Where is returning a bike difficult?

Toronto dock availability | September 16 - October 17, 2026 | Analysis version 1.0.0

Decision: identify observed return-pressure locations and investigate nearby alternatives. The evidence supports a targeted exploratory analysis, not a citywide service guarantee.

94.1%
Observed dock availability

3.4%
Successful poll coverage

11
Dates with eligible reports

What the evidence says

Coverage is the main constraint.

1,503 successful minute slots out of 44,525 expected. 21 local dates have no attempted polls. Large report counts do not compensate for missing time.

Availability varies across observed stations.

840,283 fresh, unique, operational reports cover 1,075 stations. The highest observed full rate among supported stations is 56.6% at Queen St E / Hammersmith Ave.

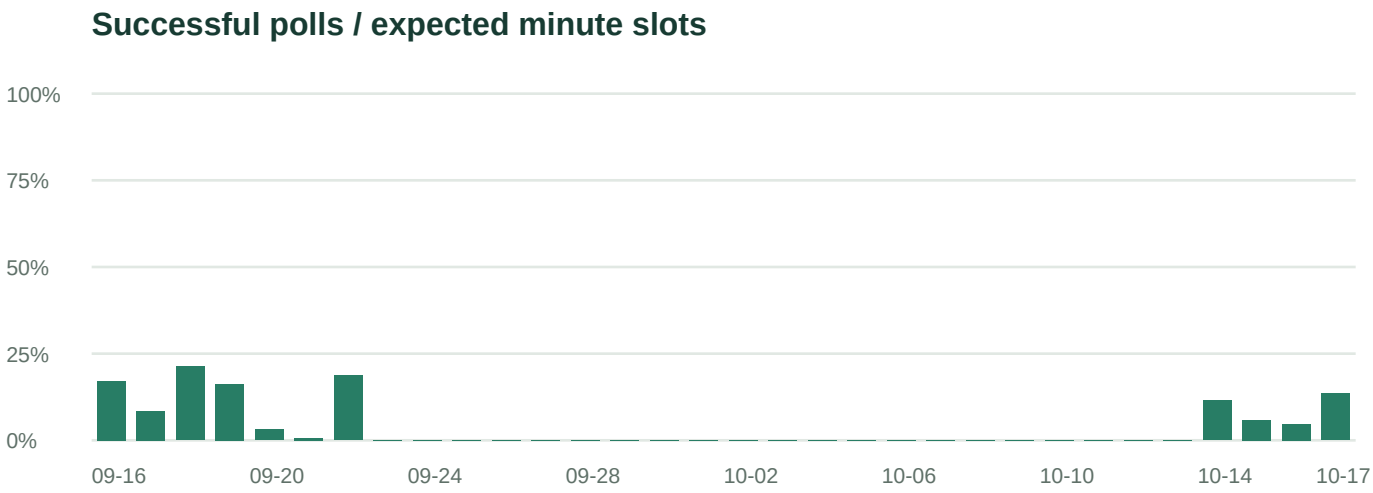
Nearby alternatives deserve investigation.

Same-feed neighboring stations are compared while selected stations are full. This describes a simultaneous option, not the chance of finding a dock after traveling there.

Recommended action: improve collector continuity first. Use live freshness-aware recommendations for actual trips; use this case study to prioritize which locations and hours to monitor next.

Coverage before conclusions

Collection completeness and report-level quality are separate measures.



The chart includes dates with no collection. September 16 starts at 14:55 and October 17 ends at 13:00 Toronto time. All other dates use complete local-day denominators. A successful poll is a collector outcome, not a rider outcome.

Quality stage or diagnostic	Records
Retrieved live snapshots	1,628,401
Unique canonical station timestamps	840,383
Repeated canonical timestamps removed	787,957
Unique reports outside freshness window	100
Eligible fresh operational reports	840,283
Conflicting station/report timestamps (diagnostic)	31,070
Quarantined records: future last_reported	198,740

Excluding conflicting timestamps gives 94.0% availability over 809,223 reports. Quarantined future clocks and conflicts warrant investigation; their causes are not established. Diagnostic counts are not an additive waterfall.

Where full docks were observed

Stations are compared only within the available collection periods.

The following stations have the largest observed full-report shares among stations with at least **100 eligible reports on 5 distinct dates**. This support rule is fixed for this exploratory release; it does not establish long-term reliability.

Station	ID	Reports	Dates	Full share
Queen St E / Hammersmith Ave	7315	782	11	56.6%
Ellis Ave / The Queensway	7478	781	11	53.3%
2300 Lakeshore Blvd W	7471	788	11	51.3%
Superior Ave / Lake Shore Blvd W	7516	800	11	49.4%
Eastern Ave / Winnifred Ave	7341	790	11	48.9%
Lake Shore Blvd W / Ellis Ave	7220	786	11	47.7%

How to read the ranking

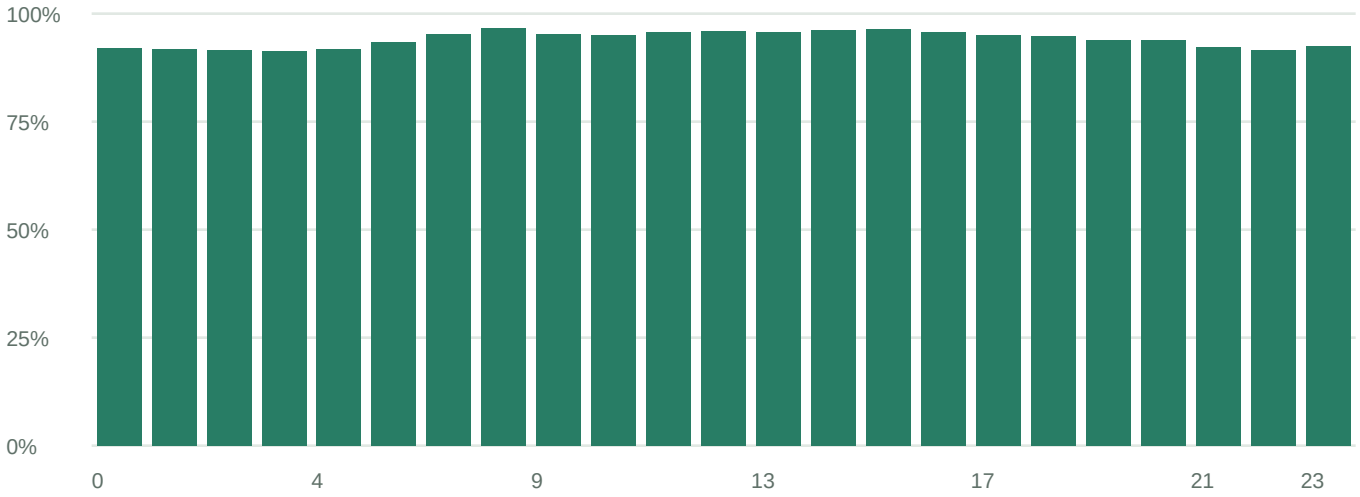
Full share = unique fresh operational reports with zero reported docks / all eligible reports for that station. The name is the last observed eligible name in the study window. Coordinates and capacity are joined through the metadata version stored with each snapshot.

1,067 of 1,075 stations meet the support rule. All stations remain in the workbook, including low-support rows. Compare report counts and observed dates before interpreting small differences. Do not call this a ranking of Toronto's worst stations or infer unmet trip demand.

When, and what is nearby?

Patterns in observations suggest follow-up questions, not causal explanations.

Observed availability by Toronto local hour



Hours pool the reports actually collected, with different dates and station mixes. The workbook separates weekdays/weekends and shows report counts and distinct dates. The chart omits hours with fewer than 100 reports or 3 observed dates; all cells remain in the workbook.

Nearest same-feed alternative for selected full stations

Station ID	Full reports	Matched	With dock	Share of matched	Mean metres
7315	443	442	219	49.5%	255
7478	416	416	262	63.0%	313
7471	404	404	400	99.0%	352

The nearest fresh operational neighbor within 750 m is selected independently of its dock count. Distance is straight-line, not a route. Unmatched reports are disclosed. The alternative may change across observations. Neither travel delay nor dock reservations are modeled.

Sensitivity: a 60-second freshness limit yields 94.1% availability over 679,157 reports, versus 94.1% over 840,283 at 300 seconds. Freshness sensitivity does not remove collection-selection bias.

Methods, limits and next steps

A reproducible analytical foundation, with a clearly bounded interpretation.

Reproduce the evidence

The repository contains versioned SQL, configuration, aggregate CSV/JSON exports, an extraction manifest with hashes, automated reconciliation tests, the Excel builder and this PDF builder. The SQL runs in a repeatable-read, read-only transaction. The frozen exports let readers rebuild the artifacts without private database access.

What this analysis does not establish

The collector was not continuously online. Missing time is nonrandom and cannot be repaired by interpolation. One station report timestamp is counted once; this can weight stations with more frequent source updates more heavily. Reported inventory changes do not identify individual pickups, returns or rebalancing. Generic bicycle compatibility is assumed. There is no seasonal generalization, live rider trial, validated forecast, or measured 82% / +14 pp result.

Recommended next steps

1. Collection: maintain continuous operation and monitor daily completeness before drawing broader patterns.
2. Analysis: repeat station/hour comparisons on well-covered, matched dates; assess stability across weeks.
3. Evaluation: freeze a historical request-replay protocol and compare simple policies before training a forecast model.

Sources and attribution

Self-collected Bike Share Toronto GBFS v3 station-information and station-status observations. Source clocks are distinct from retrieval and ingestion clocks. Raw data and credentials are not published.

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Repository and reproducible analysis

GBFS v3 specification (accessed 2026-10-17)

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